MIL-P-14769(MU)

1 Nov 1968

SUPERSEDING
PA-PD-2537A
6 July 1966

MILITARY SPECIFICATION

PROPELLANT M30 TYPE I FOR USE IN CHARGE: XM72, ZONE III

1. SCOPE

1.1 This specification covers special requirements, and provides the quality assurance provisions for the fabrication, weight to be used in charge XM72, Zone III and the preparation for delivery for one type of propellant used in charge XM72 and designated as Propellant M30 Type I.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-14766(MU) - Charge, Propellant: XM72 for 155mm
MIL-S-14770(MU) - Shell, test, Velocity, 155MM, T-5119 Assembly

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-652 - Propellants, Solid for Cannon Requirements and Packing.
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition

PUBLICATIONS

ORDNANCE CORPS

TECP-700-700, IP-70-70- U.S. Army Test and Evaluation Command Materiel Test Procedure.

FSC-1145

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material used complies with the specifications.

3.2 Propellant - Composition and physical tests. - The propellant shall comply with MIL-STD-652 for M30 Type I.

3.2.1 Process. - Details of manufacturing process and equipment used by the contractor shall be submitted to the procuring activity in writing prior to the commencement of manufacture. Any deviation from this manufacturing process shall be submitted to the procuring activity prior to being put into effect.

3.3 Propellant charge establishments.

3.3.1 Ballistics (See 4.3.3). - The propellant charge shall not exceed a Piezo Equivalent Pressure (P.E.P) that will produce projectile accelerations in excess of 8800"g"s between temperatures of -40 and +125 degrees fahrenheit when calculated as specified in 4.4.1.

3.3.2 Velocity (See 4.3.3.). - Resultant projectile velocity at the established charge weight yields a projectile mean range, under standard meteoroligical conditions, of at least 14050 meters when calculated as specified in 4.4.2.

3.3.3 Charges (See 4.3.3). - The charges used in ballistic testing shall be prepared and loaded in accordance with MIL-C-14766(MU).

3.3.4 Test Shell (See 4.3.3) - The test shells used in ballistic testing shall comply with MIL-S-14770(MU).

3.4 First article testing. - Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.5 Workmanship. - The best commercial practices shall be used in the formulation of the propellant furnished under this specification and in all other applicable documents. The propellant and its standard ingredients shall be protected from action of direct sunlight and acid fumes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2. First article inspection.

4.2.1. Quantity to be submitted. The contractor shall submit 2000 lbs of the propellant in accordance with Table I to the testing facility designated in the invitation for bids or request for proposal.

Table I
First article sample

Lot No	Average Web Size	Pounds
1	.073 ± .001	1000
2	.075 ± .001	1000

4.2.2. Inspection and testing to be performed. The First article sample shall be tested at an authorized Government Agency in accordance with the following firing program using T-5119 test shells and XM72, Zone 3 charges at 70 ± 5°F in two new M126 cannons mounted on M109 Howitzer. Chamber pressure measurements with M11 copper crusher gages and velocity measurements shall be in accordance with TECP-700-700, 1P-70-70. M107 projectiles shall be used as warming rounds in these tests.

Firing Program

No Rds.	Proj type	Zone III Charge	Purpose of Rd.	Remarks
3	M107	XM72	Warming	Est. 90% of service charge Wt.
1	T5119	XM72	Conditioning	Est. 90% of service charge Wt.
2	T5119	XM72	Curve point	Est. 90% of service charge Wt.
2	T5119	XM72	Curve point	Est. 105% of service charge Wt.
2	T5119	XM72	Curve point	Est. 100% of service charge Wt.

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Calculate Service Charge Weight

No Rds.	Proj type	Zone III Charge	Purpose of Rd.	Remarks
4	T5119	XM72	Uniformity	Calculated service charge weight

4.2.3. Acceptance of first article sample. The firing data shall be submitted to Commanding Officer, Picatinny Arsenal, Dover, New Jersey, 07801, ATTN: SMUPA-TP33. Picatinny Arsenal will determine the web size to be manufactured in the production lot.

4.3. Product and acceptance inspection.

4.3.1. Lot formation. - A lot shall consist of one or more batches of M30 Propellant produced by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of M30 Propellant which has been subjected to the same unit chemical or physical mixing process intended to make the final product homogenous.

4.3.2. Acceptance Inspection.

4.3.2.1. Materials. - The materials used in the fabrication of the propellant and charges shall comply with 3.1. Any deviation from this requirement shall reject the lot.

4.3.2.2. Propellant (See 3.2). - The propellant shall comply with 3.2. Any deviation from this requirement shall reject the lot.

4.3.2.3. Charges (See 3.3.3). - The charges furnished for ballistic testing shall comply with 3.3.3. Any deviation from this requirement shall reject the lot.

4.3.2.4. Test Shells (See 3.3.4). The test shells furnished for ballistic testing shall comply with 3.3.4. Any deviation from this requirement shall reject the lot.

4.3.2.5. Workmanship (See 3.5). - Workmanship shall comply with 3.5. Any deviation from this requirement shall reject the lot.

4.3.2.6. Surveillance test sample. - A sample shall be submitted by the contractor as specified in 4.4 MIL-STD-652.

4.3.3. Ballistic testing. - Ballistic testing will be performed by an authorized Government agency in accordance with 4.4.3 to assure compliance with 3.3. The test will determine the weight and suitability of the propellant for use in the XM-72 charge.

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4.4. Test methods and procedures. - The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerance of the test equipment used.

4.4.1. Ballistics (See 3.3.1). - The maximum "g" level at which the charge weight is to be established will be calculated by Development and Proof Services (D&PS) Aberdeen, Maryland as follows:

$$G = \frac{(P_B^* + 3 \text{ sigma}) A}{W_s + \frac{W_c}{2}} = \frac{(P_B^* + 2100)\, 29.82}{119.14 + \frac{W_c}{2}}$$

Where
P_B^* = Mean piezo equivalent pressure in breech (125 degrees fahrenheit) based on a 119.14 lb. projectile.
3 sigma = 2100 lbs. per square inch (p.s.i) (Round to round, occasion to occasion, tube to tube)
Ws = Mean weight of projectile (119.14 lbs.)
Wc = Weight of charge
G = "g" level
A = Bore area

In the above calculations the mean piezo equivalent breech pressure will be the mean of all uniformity groups, fired at estimated service charge weight and at 125 degree fahrenheit temperature, after reducing to mean projectile weight conditions and based on the pressure - charge weight curve.

4.4.2. Velocity (See 3.3.2). - Ballistics Research Laboratory (BRL) will determine the calculated service velocity level using the mean corrected muzzle velocity from the uniformity groups in 4.4.3.5.1.3. BRL will also calculate projectile mean range under standard meterological conditions.

4.4.3.0.0.0. Ballistic test (See 4.3.3.)

4.4.3.1. Test procedure. - The test procedure for standardization firings of propellants contained in TECP - 700 - 700, IP - 70 - 70, where applicable will be followed by D&PS in conducting the firing test.

4.4.3.2. Facility requirements. - Two new M126 Cannons mounted on M109 Howitzer and two M1A1 Cannons (90% min Life) mounted on M114 Howitzer will be used. Chamber pressure measurements with M11 Copper Crusher Gages and Velocity Measurements are to be in accordance with TECP - 700 - 700, IP - 70 - 70.

4.4.3.3. Ammunition. - M-107 projectiles will be used as warming rounds in these tests. T-5119 Projectiles will be used to simulate XM-454 projectile. XM72 charges with test propellant will be used for all rounds fired.

4.4.3.4. Ammunition and prefiring procedures. - The procedures outlined in TECP - 700 - 700, IP - 70 - 70 will apply. 70 and 125 degree temperatures will be employed in these tests.

4.4.3.5. Firing program.

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4.4.3.5.1. Establishment of Charge weight. - The assessed weight of charge (XM72 Zone 3) will be established by D&PS at Aberdeen Proving Ground, Aberdeen, Md. Chamber pressure data will be obtained from the following firings a at 125 degrees fahrenheit in the sequence listed.

4.4.3.5.1.1. First day.

Tube	No. Rds.	Proj. Type	Charge	Purpose of Rd.	Remarks
1	3	M107	XM72	Warming	(Est. 90% of service charge wt.)
1	1	T5119	XM72	Conditioning	(Est. 90% of service charge wt)
1	4	T5119	XM72	Curve Point	(Est. 90% of service charge wt.)

Tube	No. Rds.	Proj. Type	Charge Zone III	Purpose of Rd.	Remarks
1	7	T5119	XM72	Uniformity	(Est. 100% of service charge wt.)
1	4	T5119	XM72	Uniformity Curve Point	(105% Est. service charge wt.)

Before continuing firings, the propellant weight v s chamber pressure curve will be analysed by D&PS for necessity of a change in the estimated service weight.

Tube	No. Rds.	Proj. Type	Change Zone III	Purpose of Rd.	Remarks
2	3	M-107	XM72	Warming	(Est. 90% service charge wt.)
2	1	T5119	XM72	Conditioning	(Est. service charge wt.)
2	7	T5119	XM72	Uniformity	(Est. service charge wt.)

4.4.3.5.1.2. Second day.

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Tube	No. Rds.	Proj. Type	Charge Zone III	Purpose of Rd.	Remarks
1	3	M-107	XM72	Warming	(Est. 90% Service Charge wt.)
1	1	T5119	XM72	Conditioning	(Est. Service Charge wt.)
1	7	T5119	XM72	Uniformity	(Est. Service Charge wt.)
2	3	M107	XM72	Warming	(Est. 90% Service Charge wt.)
2	1	T5119	XM72	Conditioning	(Est. 90% Service Charge wt.)
2	4	T5119	XM72	Curve Point	(Est. 90% Service Charge wt.)
2	7	T5119	XM72	Uniformity	(Est. Service Charge wt.)
2	4	T5119	XM72	Curve Point	(105% Est. Service Charge wt.)

D&PS will determine from the above data the service charge weight which will be used for subsequent BRL plans, and shall be used for weight specification in charge XM-72 (See Dwg 8864694).

4.4.3.5.1.3. Third Day (Assessment of Service Charge Velocity). - Velocity levels in M109 and M114 Howitzers at 70 ±5 degrees fahrenheit using D&PS calculated service charge propellant weight.

How.	Tube	No. Rds.	Proj. Type	Charge Type	Zone	Purpose of Rd.	Remarks
M109	1	5	M107	M4A1	VII	Warmer	(Also Supp. data for BRL range tables)
M109	1	3	T5119	XM72	III	Uniformity	(Also supp. data for BRL range tables)
M109	2	5	M107	M4A1	VII	Warmer	(Also supp. data for BRL range tables)

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How.	Tube	No. Rds.	Proj. Type	Charge Type	Zone	Purpose of Rd.	Remarks
M109	2	3	T5119	XM72	III	Uniformity	(Also Supp. data for BRL range tables)
M114	1	5	M107	M4A1	VII	Warmer	
M114	1	3	T5119	XM72	III	Uniformity	
M114	2	5	M107	M4A1	VII	Warmer	
M114	2	3	T5119	XM72	III	Uniformity	

4.4.3.5.1.4. Fourth day. - Repeat 3 rd day Firing Program.

4.4.3.5.1.5. Fifth day. - Repeat 3 rd day Firing Program.

4.4.3.5.1.6. Sixth day. - (Assessment of Temp-Vel. Relationship)

How.	Tube	No. Rds.	Proj. Type	Charge Type	Zone	Remarks
M109	1	5	M107	M4A1		
M109	1	3	T5119	XM72		-40°F±5
M109	1	3	T5119	XM72		0°F ±5
M109	1	3	T5119	XM72		+125°F±5
M114	1	5	M107	M4A1		
M114	1	3	T5119	XM72		-40°F±5
M114	1	3	T5119	XM72		0°F ±5
M114	1	3	T5119	XM72		+125°F±5

5. PREPARATION FOR DELIVERY

5.1. Preservation, packaging, packing and marking of propellant. - Preservation, packaging and packing, and marking shall be in accordance with MIL-STD-652.

5.2. Preservation, packaging, packing, and marking of charges. - Preservation, packaging, packing and marking of charges shall be in accordance with MIL-C-14766(MU).

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5.3. Data cards. - Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1. Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Process. Details of manufacturing process and equipment used by the contractor shall be submitted to the procuring activity in writing prior to commencement of manufacture. Any deviation from this manufacturing process shall be submitted to the procuring activity prior to being put into effect.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project N. 1145-A002

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-P-14769 - Propellant M30 Type I for use in Charge XM72 Zone III

C. . AND

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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